CHAPTER 19

Logic Families (TTL, ECL, MOS, CMOS)

Bipolar and MOS technology · speed · power · fan-out · noise margin · CMOS gates · comparison

QUICK TRICK

Learn this ranking first: Speed: ECL (fastest) > TTL > CMOS. **Power:** CMOS (lowest) < TTL < ECL (highest). **Noise immunity:** CMOS (best) > TTL > ECL (worst). **Fan-out:** CMOS (highest).

19.1 What is a logic family?

● **LEVEL 1 · BEGINNER**

What is it? A **logic family** is a group of digital chips (gates, flip-flops…) built with the **same technology**. Chips of one family use the same supply voltage and the same voltage levels, so they can be joined directly.

Why do we need to know them? The technology decides how **fast** the chip is, how much **power** it uses, how many gates it can drive, and how well it resists noise.

Analogy. Car brands. Cars of one brand use the same fuel and spare parts. Different brands may need an adapter.

Technology	Main families	Built from
Bipolar	**TTL** (Transistor–Transistor Logic), **ECL** (Emitter-Coupled Logic)	Bipolar junction transistors (BJT)
MOS	**PMOS**, **NMOS**, **CMOS** (Complementary MOS)	MOS field-effect transistors (MOSFET)

EASY MEMORY

Bipolar: TTL and ECL. **MOS:** NMOS, PMOS and CMOS. CMOS = NMOS + PMOS together.

19.2 Important parameters (quick revision of Chapter 1)

● **LEVEL 2 · EXAM**

Parameter	Simple meaning	Better is…
Propagation delay t_{pd}	Time for the output to change after the input changes	Smaller
Power dissipation	Power used by one gate	Smaller
Speed-power product	t_{pd} × power (in pJ)	Smaller
Fan-out	How many gate inputs one output can drive	Larger
Noise margin	How much noise the gate can tolerate	Larger
Fan-in	Number of inputs of a gate	—

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NM_high = V_OH(min) - V_IH(min)          NM_low = V_IL(max) - V_OL(max)
Speed-power product (pJ) = t_pd (ns) x power (mW)
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Example (Level 1): standard TTL noise margin

V_{OH} = 2.4 V, V_{IH} = 2.0 V, V_{IL} = 0.8 V, V_{OL} = 0.4 V.

NM_{high} = 2.4 − 2.0 = **0.4 V**. NM_{low} = 0.8 − 0.4 = **0.4 V**.

19.3 TTL (Transistor–Transistor Logic)

● LEVEL 2 · EXAM

What is it? The most popular bipolar family (the 74xx series). A TTL gate uses **bipolar transistors**. The input transistor has several emitters (one for each input). The output stage is a **totem pole** (two transistors stacked: one pulls the output up, the other pulls it down).

Item	Standard TTL (74)
Supply voltage	+5 V
Logic levels	LOW: 0–0.8 V. HIGH: 2.0–5 V (V_{OL} = 0.4 V, V_{OH} = 2.4 V)
Noise margin	0.4 V
Fan-out	About 10
Propagation delay	About 10 ns
Power per gate	About 10 mW

TTL versions

Version	Idea	Effect
Standard TTL (74)	Basic design	10 ns, 10 mW
Schottky TTL (74S)	A Schottky diode stops the transistor from saturating	Faster (about 3 ns) but more power
Low-power Schottky (74LS)	Larger resistors + Schottky diodes	Same speed as 74 but much lower power (about 2 mW). Very popular.
Advanced (74AS, 74ALS, 74F)	Improved Schottky designs	Faster and/or lower power

QUICK TRICK

Totem pole vs open collector: A **totem-pole** output actively pulls both up and down (fast). An **open-collector** output can only pull down. It needs an **external pull-up resistor**. **Advantage:** open-collector outputs can be **tied together** to make a **wired-AND** (the output is 1 only if all gates are 1).

GATE POINT

Tri-state output: a third state, **high impedance (Z)**, like a disconnected wire. It lets many outputs share one bus: only the enabled gate drives the bus.

EASY MEMORY

Unused TTL input: an unconnected input behaves like a logic 1, but it picks up noise. Always tie it to a fixed level (for a NAND tie it to 1).

19.4 ECL (Emitter-Coupled Logic)

● LEVEL 2 · EXAM

What is it? The **fastest** logic family. Its transistors are used as a **current switch** (a differential pair). Because the transistors are **never driven into saturation**, they switch very quickly.

Item	ECL
Propagation delay	About 1–2 ns (fastest)
Power per gate	About 25 mW (high, because the current is always flowing)
Speed-power product	About 50 pJ
Logic levels	About −0.9 V (HIGH) and −1.75 V (LOW) with a supply of −5.2 V
Noise margin	Small, about 0.2 V (the worst)
Fan-out	High (about 25)
Outputs	**Complementary**: OR and NOR at the same time. Outputs can be tied for a **wired-OR**.
Used in	High-speed systems: supercomputers, communication

QUICK TRICK

ECL: fastest, but high power and low noise margin. It is fast because the transistors never saturate (no storage delay).

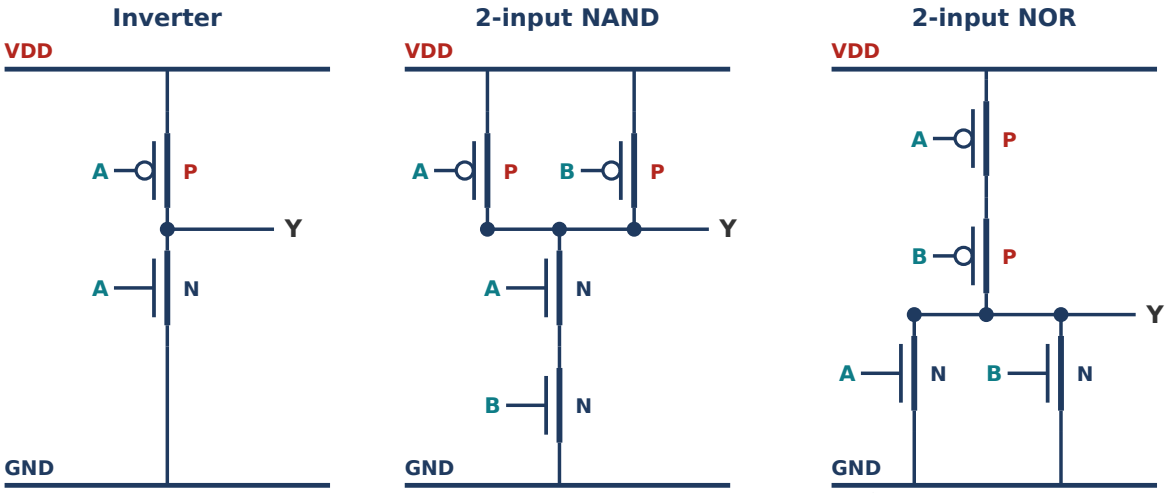
19.5 MOS and CMOS

● LEVEL 2 · EXAM

MOSFETs are controlled by the voltage on the gate (no steady gate current). There are two types. **Remember: NMOS is ON when its gate is 1. PMOS is ON when its gate is 0.**

	NMOS transistor	PMOS transistor
Conducts (ON) when gate is	**1 (HIGH)**	**0 (LOW)**
Used in CMOS as	**Pull-down** (connects the output to ground)	**Pull-up** (connects the output to V_{DD})
Symbol	Gate with no bubble	Gate with a bubble

CMOS idea. Use a PMOS network (pull-up) and an NMOS network (pull-down) together. For any input, **exactly one network is ON**. So the output is either connected to V_{DD} or to ground, and **almost no current flows from V_{DD} to ground** when nothing changes.



A	B	NAND: PMOS-A, PMOS-B, NMOS-A, NMOS-B	NAND output
0	0	ON, ON, OFF, OFF	1
0	1	ON, OFF, OFF, ON	1
1	0	OFF, ON, ON, OFF	1
1	1	OFF, OFF, ON, ON	0

QUICK TRICK

CMOS gate rule: NAND = NMOS in series, PMOS in parallel. NOR = NMOS in parallel, PMOS in series. The PMOS network is always the **dual** (opposite) of the NMOS network. *Why:* series needs **all** gates to be 1 (AND-like), so series NMOS pulls the output down only when A AND B are 1: that is a NAND.

GATE POINT

Transistor count: inverter = 2, 2-input NAND or NOR = 4, **n-input NAND or NOR = 2n**. A 2-input AND = NAND + inverter = **6** transistors.

Why CMOS uses so little power

In a steady state one network is OFF, so there is **no path from V_{DD} to ground**: static power is almost zero. Power is used mainly when the output **changes**, to charge and discharge the load capacitance C:

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Dynamic power   P = C x V_DD^2 x f
(C = load capacitance, f = switching frequency)
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Example (Level 3): CMOS dynamic power

Given: C = 20 pF, V_{DD} = 5 V, f = 2 MHz.

Concept: $P = C V_{DD}^2 f$.

Calculation: $20 \times 10^{-12} \times 25 \times 2 \times 10^{6} = 1 \times 10^{-3}$ W.

Answer: 1 mW. (Doubling the frequency or the supply voltage squared changes the power: $P \propto f$ and $P \propto V^2$.)

COMMON MISTAKE

A CMOS gate that is **not switching** uses almost no power, but a CMOS gate switching at high frequency can use **more** power than expected. CMOS power depends on frequency. Also: **never leave a CMOS input unconnected** (it floats, picks up noise and wastes power). Tie it to V_{DD} or ground.

CMOS strong points	CMOS weak points
Very low static power	Slower than bipolar (older series)
High noise margin (about 30–45 % of V_{DD})	Easily damaged by static electricity (ESD)
Wide supply range (3–15 V for the 4000 series)	Power rises with frequency
High fan-out (50 or more, limited by capacitance)	Possible latch-up
Small and easy to integrate (high density)	

EASY MEMORY
PMOS chips were the first MOS chips (slow). **NMOS** is faster and denser (used in memories and early microprocessors) but uses power when the output is LOW. **CMOS** is the choice for nearly all modern digital chips.

19.6 Comparison of the families

● **LEVEL 2 · EXAM**

Parameter	Standard TTL	74LS (TTL)	ECL	CMOS
Propagation delay	~10 ns	~10 ns	**~1–2 ns (best)**	~10–50 ns (series dependent)
Power per gate	~10 mW	~2 mW	~25 mW (highest)	**~0.01 mW static (lowest)**
Speed-power product	~100 pJ	~20 pJ	~50 pJ	Very low at low frequency
Fan-out	~10	~20	~25	**50 or more**
Noise margin	~0.4 V	~0.4 V	~0.2 V (worst)	**Best** (30–45 % of V_{DD})
Supply	+5 V	+5 V	−5.2 V	3–15 V (wide)

EASY MEMORY
These numbers are typical values and differ a little between books. **Learn the ranking, not the exact values.**

Interfacing. A TTL HIGH output (about 2.4 V) is too low for a 5 V CMOS input (needs about 3.5 V). Add a **pull-up resistor** or use **HCT** chips (CMOS chips with TTL-compatible inputs). A CMOS output can usually drive TTL inputs.

Quick-Solve Tricks — Summary

Topic	Trick
Which is fastest?	ECL (transistors never saturate).
Lowest power / best noise margin / highest fan-out	CMOS.
Highest power	ECL.
NMOS / PMOS	NMOS ON when gate = 1. PMOS ON when gate = 0.
CMOS NAND / NOR	NAND: NMOS series, PMOS parallel. NOR: NMOS parallel, PMOS series.

Topic	Trick
Transistors	n-input NAND or NOR = 2n. AND = 6.
CMOS power	$P = C V^2 f$ (depends on frequency).
Open collector	Needs pull-up. Tied outputs = wired-AND.
Tri-state	0, 1 and high impedance (Z). Used on buses.
Unused inputs	TTL floats to 1 (tie anyway). CMOS: must be tied.

GATE QUICK CHECK — Chapter 19

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ECL fastest.  CMOS lowest power, best noise margin, highest fan-out.
TTL in between.
NM_H = V_OH - V_IH      NM_L = V_IL - V_OL       SPP = t_pd x P
CMOS power  P = C V_DD^2 f        n-input CMOS NAND/NOR = 2n transistors
```

Traps: CMOS is not always the fastest · CMOS power depends on frequency · NAND has NMOS in series (not parallel) · open-collector outputs need a pull-up · never leave CMOS inputs floating.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Which logic family is the fastest?

Q2. Which logic family has the lowest static power?

Q3. Does an NMOS transistor conduct when its gate is 0 or 1?

● **LEVEL 2 · EXAM**

Q4. V_{OH} = 2.4 V, V_{IH} = 2.0 V, V_{IL} = 0.8 V, V_{OL} = 0.4 V. Find both noise margins.

Q5. A gate has t_{pd} = 10 ns and uses 10 mW. Find its speed-power product.

Q6. How many transistors are in a 2-input CMOS NAND gate? In a 3-input CMOS NOR gate?

● **LEVEL 3 · GATE**

Q7. A CMOS gate drives C = 20 pF at f = 2 MHz with V_{DD} = 5 V. Find the dynamic power.

Q8. How are the transistors connected in a 2-input CMOS NOR gate?

Q9. The open-collector outputs of three TTL gates are tied together with one pull-up resistor. What logic function does the common output give?

Q10. What is the minimum number of transistors for a 2-input CMOS AND gate?

Q11. A TTL gate has V_{OH} = 2.4 V and drives a 5 V CMOS input that needs V_{IH} = 3.5 V. Will it work reliably? What is the fix?

Q12. In a 2-input CMOS NAND gate, A = 1 and B = 0. Which transistors are ON? What is the output?

Answers

Q	Solution
1	**ECL** (its transistors do not saturate).
2	**CMOS**.
3	**1**. An NMOS transistor is ON when the gate is HIGH.

Q	Solution
4	$NM_{high} = 2.4 - 2.0 =$ **0.4 V**. $NM_{low} = 0.8 - 0.4 =$ **0.4 V**.
5	SPP = 10 ns × 10 mW = **100 pJ**.
6	2n transistors: 2-input NAND = **4**. 3-input NOR = 2 × 3 = **6**.
7	$P = C V^2 f = 20 \times 10^{-12} \times 25 \times 2 \times 10^6 =$ **1 mW**.
8	**PMOS in series** (pull-up) and **NMOS in parallel** (pull-down).
9	**Wired-AND**. The output is 1 only when all three gates give 1 (any gate pulling down gives 0).
10	NAND (4) + inverter (2) = **6** transistors.
11	**No.** 2.4 V is below the required 3.5 V. Fix: add a pull-up resistor to +5 V, or use a TTL-compatible CMOS family (HCT).
12	A = 1: PMOS-A is OFF, NMOS-A is ON. B = 0: PMOS-B is ON, NMOS-B is OFF. The series NMOS path is broken, the PMOS-B path connects the output to V_{DD}. **Output = 1**.

Next: Chapter 20 — GATE Quick Revision (all formulas, tricks and traps on a few pages). Then Chapter 21 — a full practice set with answers.

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